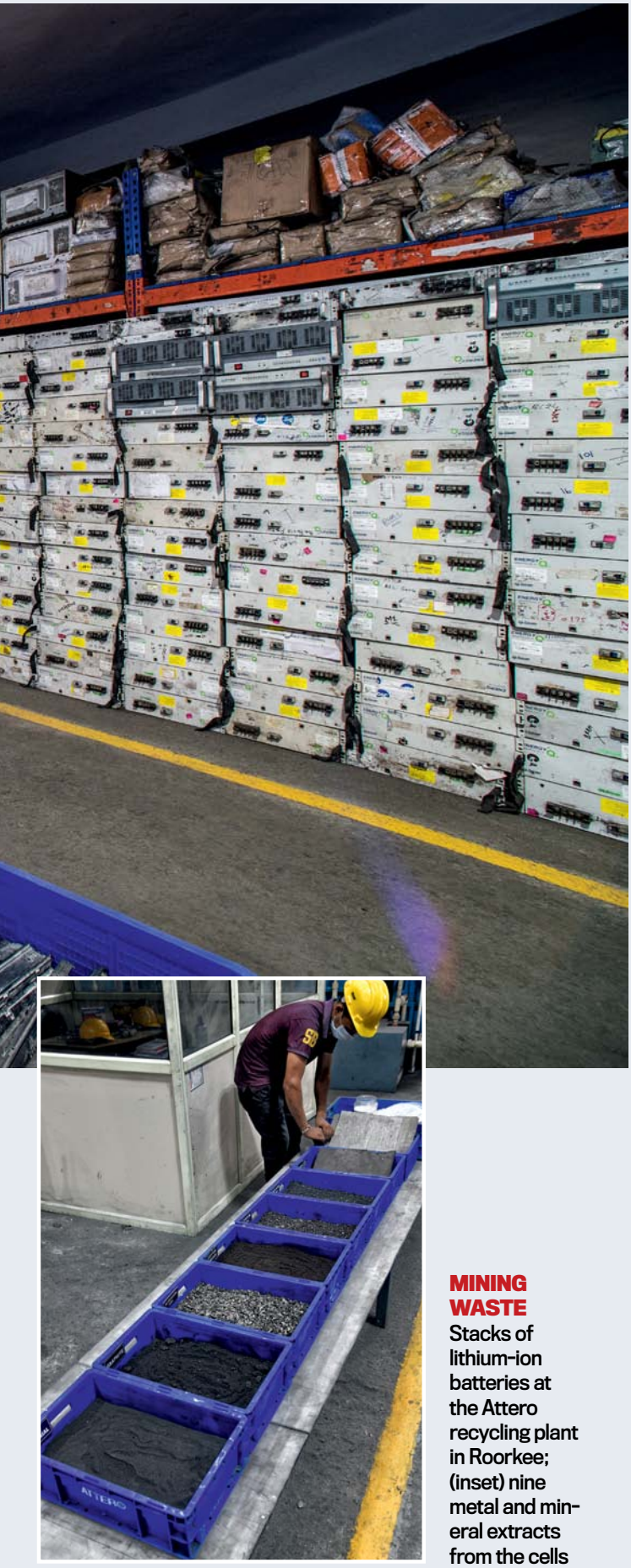


MINING WASTE Stacks of lithium-ion batteries at the Attero recycling plant in Roorkee; (inset) nine metal and mineral extracts from the cells

ally come down much, which means these processes still continue," says Priti Mahesh, chief programme coordinator of Delhi-based environmental NGO Toxics Link, which has been working on e-waste for two decades. "There are a lot of primary studies outside India that show the kind of impact e-waste can have. So, we know the kind of hazard and environmental pollution e-waste ends up causing." Electronic waste recycling involves handling heavy metals such as mercury, lead and cadmium besides organic pollutants such as dioxins, and other air pollutants emitted during burning. A 2021 World Health Organization report identified more than 1,000 harmful substances as being either components of e-waste or artisanal processing systems.

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The community of e-waste recyclers mostly comes from lower-income groups, employing cheap labour and hazardous methods to extract metals. They cannot be bothered with proper protective gear or techniques. The crude extraction of gold is a fairly old technique involving leaching using acids, exposing the worker to burns. Polychlorinated biphenyls—under phaseout and found in capacitors and transformers—are known to be highly toxic, impacting the immune and reproductive systems and possibly even causing cancer. Open burning—of circuit boards and polyvinyl chloride-insulated copper cables—releases pollutants, causing respiratory damage. That informal disposal of e-waste contaminates water, soil and air and eventually reaches our food streams is well known. The 2021 WHO report cited recent studies of soil samples from four e-waste recycling spots in India—New Delhi, Kolkata, Mumbai and Chennai—suggesting 'an association between informal recycling and elevated levels of toxic PCBs in soils in surrounding metropolitan areas'.